

Curriculum Vitae (CV)

NAYUNI VANDANA

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ACADEMIC QUALIFICATION

Indian Institute of Technology Tirupati, India

Ph.D., Chemical Engineering, 2023-2027 (% /CGPA - 9.0/10) (Ongoing)

- Project title: “*Membrane Processes for Valorization of Whey and Cold Sterilization of Liquid Foods*”
- Supervisors: Prof. Anki Reddy Katha and Prof. KSMS Raghavarao

Tamil Nadu Veterinary and Animal Sciences University, College of Food and Dairy Technology, Tamil Nadu

M.Tech, Dairy Chemistry, March 2021-2023 ((% /CGPA - 9.28/10; First Division with Distinction)

- Project title: “*Shelf-life Extension of Paneer by Edible Coatings Using Whey Protein Concentrate and Natural Deep Eutectic Solvents*”
- Supervisor: Dr. Esther Magdalene Sharon

Sri Venkateswara Veterinary University, College of Dairy Technology, Tirupati

B. Tech., Dairy Technology, 2016-2020 (% /CGPA – 8.099/10; First Division with Distinction)

- Project title: “*Design of Shell and Tube Heat Exchanger*”
- Supervisor: Dr. J. Rajsekhar

TECHNICAL PROFICIENCY

Software Tools	MS Office, Origin Pro, ANOVA, SPSS, Design Expert Software, and general internet tools.
Instruments	Membrane Stirred Cell, Pilot Scale Membrane Setup, UV-Visible spectrophotometer, Film Applicator (Membrane Casting), Electrospinning, Hunter lab calorimetry, Texture Profile Analyzer, Kjeldahl Apparatus, Digital Compound Microscope, Biosafety Cabinet, Spray Dryer, Tray Dryer, Rotary Vacuum Evaporator and Vacuum Packaging Machine Etc.

INDUSTRIAL INTERSHIPS

- **Heritage Foods Pvt Limited, Gokul Plant, Kasipentla, Tirupati** (16.07.2019 – 05.11.2019)
- **Sangam Milk Producer Company Limited Vadamudi, Guntur** (06.11.2019 to 19.11.2019)
- **Mother Dairy Fruits and Vegetables Private Limited, Avilala, Tirupati** (10.02.2022 – 26.03.2022)

ACADEMIC EXPERIENCE

Half-time Teaching Assistant Duties (Dept of Chemical Engineering, IIT Tirupati)

1. Microscale Unit Operations, Ecology and Environment, Fluid and Particle Mechanics and Transport Processes: **Prof. Anki Reddy Katha**
2. Life Sciences: **Dr. S. Uday Kumar**
3. Separation Processes: **Dr. Narendra Singh and Prof. Anki Reddy Katha**

TRAININGS & WORKSHOPS

1. Participated in the “*Advances in Membrane Science and Separation Technologies*” Course organised by the Dept of Polymer and Process Engineering under the aegis of the continuing education centre at the **Indian Institute of Technology Roorkee**, from 28th July 2025 to 2nd August 2025.
2. **Central Salt & Marine Chemicals Research Institute, Bhavnagar** (CSIR CSMCRI), Bhavnagar, Gujarat – “*Membrane Fabrication.*”

3. Participated in the National Seminar (online) on “**Microplastics’ Removal from Environmental Matrices: Current Scenario, Challenges, & Future Perspectives**”, held at ICMR -NIREH, Bhopal during August 22 – 23, 2024.
4. Awarded **Elite** NPTEL online certification on “**Novel Technologies for Food Processing and Shelf-Life Extension**” from **Indian Institute of Technology Kharagpur** funded by the MOE, Govt. of India.
5. Training under **SERB Sponsored: High-End Workshop** on “**Recent Trends in Sensing Techniques for Food Quality Evaluation**” at the National Institute of Food Technology, Entrepreneur and Management (NIFTEM-T) Thanjavur, Tamil Nadu.

PUBLICATIONS

1. **Nayuni Vandana**, KSMS Raghavarao and K. Anki Reddy. Membrane Processing in the Dairy Industry: Challenges and Future Perspectives (*Under review*).
2. **Nayuni Vandana**, Esther Magdalene Sharon M., Sivakumar G M., and Geetha P. (2022). Optimization of Whey Protein Concentrate-Natural Deep Eutectic Solvents Based Edible Coating of Paneer: *Biological Forum – An International Journal*, 14 (4a): 551-558.
3. **Nayuni Vandana**, Pavan Kumar Reddy. C and Sangita Ganguly (2024). Development of Probiotic Processed Cheese Spread: *Journal of Novel Research and Innovative Development*. Volume 2 Issue 8. (Doi: <http://doi.one/10.1729/Journal.41258>).
4. **Nayuni Vandana**, Pavan Kumar Reddy. C and Sangita Ganguly (2024). Functional Milk Proteins Based Edible Coating of Processed Cheese: *Journal of Novel Research and Innovative Development*. Volume 2 Issue 8. (Doi: <http://doi.one/10.1729/Journal.41259>).

TECHNICAL/ POPULAR ARTICLES

1. **Nayuni Vandana** and I. Mohanakkaviya. Novel Technologies for Membrane Processes. *Agriculture and Food: e-Newsletter*, 5(7), 2023. ISSN:2581 8317. (Article ID: 43105)
2. **Nayuni Vandana** and I. Mohanakkaviya. Functional Milk Proteins as a Biopolymers. *Agriculture and Food: e-Newsletter*, 5(9), 2023. ISSN:2581 8317. (Article ID: 43834)
3. Mohanakkaviya and **Nayuni Vandana**. An outline of Electrohydrodynamic Techniques. *Agriculture and Food: e-Newsletter*, 5(7). 2023. ISSN:2581 8317. (Article ID: 43099)
4. Mohanakkaviya and **Nayuni Vandana**. Refractive Window Drying – An Emerging Drying Technique. *Agriculture and Food: e-Newsletter*, 5(9). 2023. ISSN:2581 8317. (Article ID: 43835)

INTERNATIONAL CONFERENCES/ SYMPOSIUM

1. Presented a **POSTER** titled entitled “**Comparative Performance of Ultrafiltration Membranes for Dairy Whey Protein Separation**” at the International Conference on Membranes for Decarbonization and Sustainability (MDCS) & 2nd Conference on Applied Membrane Technologies and Translational Research (AMTTR-2), jointly organized by the Indian Membrane Society (IMS) and Gogri Hub for Membrane Research, IIT Bombay (**Best Poster Award**).
2. Presented a **PAPER** entitled “**Functional Milk Proteins Based Edible Coating of Processed Cheese**” at the **6th International Conference on Food & Nutrition**, organized by the Universal Society of Food and Nutrition (USDN) on 25th and 26th October 2023 via Virtual mode in Canada. (**Best E-Oral Presentation**)
3. Presented a **POSTER** titled “**Effect of Whey Protein Concentrate-Natural Deep Eutectic Solvents Based Edible Coating of Paneer**” at the Hybrid **International Conference on Innovations in Food and Beverage Technology for Sustainable Future (IFBT)**, Organized by the Tamil Nadu Veterinary and Animal Sciences University, College of Food and Dairy Technology, Chennai, from 4th to 6th January 2023. (**Best Poster Presentation - 3rd Prize**)
4. Participated in an **Indo-Canadian Symposium** on Water Management: **Sustainability & Impact of Climate Change** (2023). IIT Tirupati and Dalhousie University.
5. Participated in an **Indo-Canadian Symposium** on Water Management: **Adaptation to Climate Change and Sustainability**, held online on February 22 & 23, 2024, at IIT Tirupati, funded by the Shastri Conference & Lecture Series Grant by Shastri Indo-Canadian Institute.

NATIONAL CONFERENCES

1. Presented a POSTER titled “***Shelf-Life Extension of Paneer by Edible Coatings Using Whey Protein Concentrate and Natural Deep Eutectic Solvents***” in the **49th Dairy Industry Conference on India to the World: Opportunities & Challenges** Organized by the Indian Dairy Association and Gujarat State Chapter from March 16th - 18th March 2023 at Helipad Exhibition Centre, Gandhinagar, Gujarat.
2. Presented a POSTER entitled “***Development of Probiotic Processed Cheese Spread***” at the **27th Indian Convention of Food Scientists and Technologists** (27th ICFOST 2020) organized by the AFSTI, CSIR-CFTRI, DRDO – DFRL and Tezpur University, Assam.

HONOURS & AWARDS

1. Awarded **Best E-Oral Presentation** in “**6th International Conference on Food & Nutrition**” organized by Universal Society of Food and Nutrition (USDN) on 25th and 26th October 2023 via Virtual mode in Canada.
2. Nominated in the “**16th National Dairy Products Judging Contest - 2019**” organized by SMC College of Dairy Science & Alumni Association, AAU, Anand, Gujarat on September 07, 2019.
3. **First prize in 2D and 3D Exhibits Model** for National Milk Day Celebration, 2021 organized by the College of Food and Dairy Technology, Tamil Nadu.
4. Achieved **1st prize in English Essay Writing** in the College Day Celebrations of College of Dairy Technology, Tirupati during the year 2017-18.
5. Achieved **1st prize** in the **New Innovative Model Project Preparation** competition conducted during National Technology Day- 2022 at the College of Food and Dairy Technology, Chennai.
6. Achieved **1st prize** in an **E-Poster competition** for World Food Day celebrations on the theme “*Leave No One Behind*” organized by the College of Food and Dairy Technology, Chennai.
7. Achieved **2nd Prize** in an **English Essay Writing** competition held eve of Mahatma Jyoti Rao Phule birthday celebration on 11th April 2017.
8. Awarded **2nd prize** in “**Dharma Praveshika Vignan pariksha**” in 2012 conducted by Tirumala Tirupati Devasthanam (TTD)- Hindu Dharma Pracharaparishat, Tirupati.
9. Achieved **2nd prize** in **solo singing** at the Annual Sports, Games, Cultural and Literary events conducted by Sri Venkateswara Veterinary University, College of Dairy Technology, Tirupati from 2017 to 2018.
10. Awarded **3rd prize** in a Hybrid International Conference on **Innovations in Food and Beverage Technology for Sustainable Future (IFBT)** from 4th to 6th January 2023 organized by the College of Food and Dairy Technology, Chennai.
11. Achieved **3rd prize** in **solo singing** at the Annual Sports, Games, Cultural and Literary events conducted by Sri Venkateswara Veterinary University, College of Dairy Technology, Tirupati during the year 2018-2019.
12. **Third prize** for **BANQUET WITHOUT FIRE** Competition held in correspondence to National Milk Day Celebration-2021.
13. Secured **3rd prize** in **Group Dance** in the college day celebrations of College of Dairy Technology, Tirupati during 2017-2018.

POSITIONS & RESPONSIBILITIES

1. Served as a **Volunteer** for the **National Service Scheme** on a **Special NSS Camp** from 11.03.2019 to 17.03.2019 at Reddy Vari Palli organized by Sri Venkateswara Veterinary University, Tirupati.
2. **Student Committee Member** in the National conference on “**Challenges and Opportunities for The New Generation Dairy Foods in India**” jointly organized by the College of Dairy Technology and Sri Venkateswara Veterinary University, Tirupati- 5175079 (AP) and National Academy of Dairy Science (India), Karnal, Haryana during 9th and 10th September 2018.
3. **Leadership Role** for the **Student Ready Rural Development Program** at Sri Venkateswara Veterinary University, College of Dairy Technology, Tirupati.
4. **Team leader** for Student READY Experiential Learning Module at Sri Venkateswara Veterinary University, College of Dairy Technology, Tirupati.
5. Participated in **cocurricular activities** ranging from literary, sports, artistic competitions and social activities.